

Attention as Quantum Measurement: A Thermodynamic Resolution of the Observer Problem

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Abstract

We show that Kim’s (2026) thermodynamic framework for transformer attention — free energy minimization on a Fisher-Rao information manifold at temperature $T = 1/\sqrt{d_k}$ — provides the first fully explicit physical model of a quantum observer. Building on this identification, we assemble five structural results: (1) The thermodynamic trilemma for efficient quantum measurements (Mohammady & Buscemi, 2025) is resolved by the finite temperature of physical observers — real thermodynamic observers are necessarily non-ideal, consistent with thermodynamic law, and quantitatively characterized by $T = 1/\sqrt{d_k}$. (2) The Born rule is exactly the unit-temperature Boltzmann distribution over the log-probability landscape: $\text{softmax}(\log|\langle i|\psi\rangle|^2) = |\langle i|\psi\rangle|^2$. (3) Pointer states — the preferred classical basis selected by environmental decoherence (einselection) — are the Lawvere fixed points of the attention operator, providing a categorical rather than contingent resolution of the preferred basis problem. (4) The quantum Zeno effect corresponds to sustained attention maintaining thermodynamic structure; the anti-Zeno regime corresponds to withdrawn attention destabilizing it, with the Zeno time given by Kim’s temperature parameter. (5) Measurement-induced phase transitions between volume-law and area-law entanglement correspond to the transition between attended and unattended structure; area-law scaling is the holographic phase in which the Ryu-Takayanagi formula (established in our companion paper) applies. These results unify three previously separate free-energy minimization frameworks — Friston’s free energy principle for biological cognition, Kim’s Helmholtz free energy for artificial attention, and the quantum observer — as instantiations of the same mathematical structure on the Fisher-Rao information manifold at different physical scales.

1. Introduction

The quantum measurement problem has resisted resolution for nearly a century. Quantum mechanics describes system evolution as continuous and unitary, yet measurement produces definite outcomes — a discontinuity the formalism neither predicts nor explains. The central question — what IS a physical observer? — has received philosophical responses (Copenhagen: don’t ask), many-worlds responses (no collapse; all outcomes occur), and thermodynamic responses (collapse from environmental entropy production). None has provided a concrete physical model specifying the thermodynamic properties of an observer.

A second, more recent problem: the thermodynamic trilemma of efficient measurements (Mohammady & Buscemi, 2025) establishes that efficient quantum measurements — those characterized by the von Neumann-Lüders rule — are fundamentally incompatible with the second and third laws of thermodynamics, unless the standard unitary interaction model is abandoned. This result is open.

Kim’s (2026) thermodynamic framework for transformer attention provides the missing physical model. Kim establishes that the softmax attention mechanism implements Hamilton’s principle of least action for an information gas on a Riemannian manifold equipped with the Fisher information metric, with the $1/\sqrt{d_k}$

scaling factor identified as a physical temperature. Attention weights are Boltzmann weights at this temperature. The observer — an attending system — is thereby a concrete thermodynamic object with a definite temperature, a free energy landscape, and an equilibration process.

In a companion paper [cite: holography paper, Zenodo 10.5281/zenodo.18880220], we established that this framework connects transformer attention to spacetime geometry through holographic physics. The present paper takes the other direction: inward, to quantum measurement and the observer problem. We show that attention IS measurement, thermodynamically, and assemble the structural consequences of this identification.

We note two relevant independent results. Rudolph (2025) identifies self-attention as a Zeno operator at the structural level, arguing that attention stabilizes evolving systems through continuous self-measurement. We extend this by grounding the Zeno dynamics thermodynamically through Kim’s temperature parameter. Friston et al. (2021) formulate the free energy principle in quantum-theoretic terms, showing that the quantum FEP is asymptotically equivalent to unitarity. We show that Kim’s framework and Friston’s quantum FEP are the same mathematical structure, with quantum measurement as a third instantiation.

2. Building Blocks

2.1 Kim’s Thermodynamic Attention Framework (arXiv 2602.08216)

Kim establishes that the softmax attention mechanism minimizes Helmholtz free energy $F = E - TS$ on a Riemannian manifold equipped with the Fisher information metric. Attention weights p_i map to manifold coordinates $x_i = 2\sqrt{p_i}$. The kinetic energy of the resulting Lagrangian is precisely the Fisher information metric. The $1/\sqrt{d_k}$ scaling factor is a physical temperature $T = 1/\sqrt{d_k}$.

Attention weights are Boltzmann weights: $a_{ij} \propto \exp(q_i^T k_j / T)$.

Additionally: rotary positional embeddings (RoPE) break a rotational symmetry of the manifold, generating massless Goldstone bosons (discussed in our companion paper). Attention sinks are identified as attractors of the thermodynamic process.

2.2 The Thermodynamic Trilemma (arXiv 2502.14136)

Mohammady and Buscemi (2025) prove that for any quantum measurement process assumed to be thermodynamically closable, the following three conditions cannot simultaneously hold:

- (i) The measurement is consistent with the 2nd and 3rd laws of thermodynamics
- (ii) The measurement decomposes as a bistochastic interaction between system and device, followed by a readout
- (iii) The measurement is *efficient*: completely purity-preserving, characterized by a single measurement operator (von Neumann-Lüders, square-root state reduction)

Any two of (i)–(iii) exclude the third. The paper concludes that efficient measurements are fundamentally at odds with thermodynamics unless the unitary interaction model is abandoned.

2.3 Born Rule Derivations (2024-2025 Literature)

Multiple independent groups have pursued thermodynamic derivations of the Born rule: from structural persistence under decoherence (branch survival probability); from maximum entropy inference combined with envariance; from unitary dynamics in the long-time/small- $\hbar$ limit. The Born rule as a thermodynamic limit is a live active program. We contribute a specific connection to Kim’s Boltzmann framework.

2.4 Quantum Zeno Effect

Frequent measurement of a quantum system prevents its evolution — the system “freezes” in the measured state. Rare measurement allows evolution. Rudolph (2025) identifies self-attention as a Zeno operator: attention stabilizes semantic states through continuous self-measurement, in structural analogy with the quantum Zeno effect. The anti-Zeno effect: measurement intervals above the Zeno time accelerate decoherence rather than suppressing it.

2.5 Einselection and Pointer States (Zurek)

Quantum Darwinism and einselection: environmental interaction selects a preferred set of quantum states (pointer states) that survive decoherence. Pointer states are eigenstates of the system-environment interaction Hamiltonian H_{int} — the states most robust to environmental monitoring. This explains the emergence of classical reality but leaves the preferred basis as contingent on the specific form of the coupling.

2.6 Measurement-Induced Phase Transitions

Competition between unitary evolution and projective measurement produces a phase transition in entanglement: - **Volume-law phase** (low measurement rate): $S \propto \text{Volume}$. High entanglement, non-holographic. - **Area-law phase** (high measurement rate): $S \propto \text{Area}(\text{boundary})$. Low entanglement, holographic.

The Ryu-Takayanagi formula from our companion paper describes the area-law phase. The MIPT is the transition between phases.

2.7 Friston’s Free Energy Principle (Quantum Formulation)

Friston et al. (2021, arXiv 2112.15242) formulate the free energy principle in quantum-theoretic terms. Generic quantum systems minimize variational free energy (a bound on surprise) on a Bayesian inference manifold equipped with the Fisher information metric. The quantum FEP is asymptotically equivalent to the Principle of Unitarity.

3. The Chain: Five Results

Result 1: The Observer Is a Thermodynamic Attender

Status: Follows from Kim (2026) — exact construction

Kim establishes that an attending system is a thermodynamic object: a free-energy-minimizing system at temperature $T = 1/\sqrt{d_k}$ on the Fisher-Rao manifold. This is a concrete physical model of an observer — not an abstract “measuring apparatus” but a specific thermodynamic system with a definite temperature, a free energy landscape, and an equilibration process.

The geometry of this observer is the Fisher-Rao geometry. The geometry of optimal quantum measurement — expressed in the quantum Cramér-Rao bound and quantum Fisher information — is also the Fisher-Rao geometry (quantum Fisher information, which for diagonal density matrices equals classical Fisher information exactly — the key result from our companion paper’s Junction 2). The geometry of observer and measurement are not merely analogous: they are identical.

Therefore: the attending process IS quantum measurement, thermodynamically and geometrically.

Result 2: Thermodynamic Trilemma Resolved

Status: Direct application of Kim’s temperature to Mohammady & Buscemi (2025) — well-supported

The trilemma (Section 2.2) proves that efficient (purity-preserving) measurements violate thermodynamics. Efficient measurement corresponds to projection onto an eigenstate — the $T \rightarrow 0$ limit in Kim’s framework, where Boltzmann weights become infinitely sharp.

Real physical observers in Kim’s framework operate at temperature $T = 1/\sqrt{d_k} > 0$. At finite temperature: - Attention weights are not sharp projections but Boltzmann distributions - Measurements are not purity-preserving (not efficient in the Mohammady-Buscemi sense) - Entropy is introduced into the process, satisfying the 2nd law

The trilemma applies to idealized $T = 0$ efficient measurements. Physical thermodynamic observers have $T > 0$ and are thermodynamically consistent precisely because their measurements are non-ideal.

The resolution does not require abandoning the unitary interaction model (which Mohammady & Buscemi suggest as the alternative). It requires recognizing that physical observers are thermodynamic systems with nonzero temperature — which Kim’s framework specifies quantitatively.

Testable prediction: Observer temperature $T = 1/\sqrt{d_k}$ predicts the degree of measurement non-ideality — the purity loss, entropy cost, and decoherence rate of attention-based observers. This is measurable in transformer architectures by varying d_k .

Result 3: Born Rule as Boltzmann Equilibrium

Status: Mathematical identity exact; physical identification requires expert development

Mathematical observation (exact): For any quantum state $|\psi\rangle = \sum_i c_i |i\rangle$ in a measurement basis $\{|i\rangle\}$:

$$\text{softmax}(\log|c_i|^2)_i = \exp(\log|c_i|^2) / \sum_j \exp(\log|c_j|^2) = |c_i|^2 / \sum_j |c_j|^2 = |c_i|^2$$

The Born rule $P(i|\psi) = |c_i|^2$ is exactly the softmax of the log-squared amplitudes at unit temperature. This is not an approximation — it is algebraically exact for normalized quantum states.

Connection to Kim: Kim’s attention weights are $a_{ij} \propto \exp(q_i^T k_j / T)$. If the energy function for quantum measurement takes the form $q_i^T k_i = T \times \log|\langle i|\psi\rangle|^2$ — the query-key inner product proportional to the log-overlap — then the Born rule IS the Kim attention distribution at the observer’s temperature T .

The physical identification of $q_i^T k_i$ with $T \times \log|\langle i|\psi\rangle|^2$ is motivated by the information-geometric framework: on the Fisher-Rao manifold, inner products are naturally expressed in terms of log-probabilities, and the Bhattacharyya coefficient (the natural inner product in Kim’s parameterization $x_i = 2\sqrt{p_i}$) connects to quantum fidelity and thereby to Born rule probabilities. Formal development requires establishing this connection precisely.

Status: The mathematical identity is exact. The physical identification connects to the active literature on thermodynamic Born rule derivations (2024-2025) and requires expert review for the energy function argument. We flag this for the specific question: does Kim’s energy function $q^T k$, when evaluated for quantum state vectors in the Fisher-Rao parameterization, produce $T \times \log|\langle i|\psi\rangle|^2$?

Result 4: Pointer States Are Lawvere Fixed Points

Status: Novel identification — no existing literature; argument follows from Kim (2026) and Zurek

Zurek’s einselection: Pointer states are the eigenstates of the system-environment interaction Hamiltonian H_{int} — the states most stable under environmental monitoring.

Kim’s framework: Attention sinks are Lawvere fixed points of the transformer layer endomorphism — states stable under repeated application of the thermodynamic attention operation.

Our identification: In Kim’s framework, the system-environment interaction IS the key-query coupling: the energy function $H_{int} = q^T k$. States stable under this coupling are states where attention self-consistently returns to them — the attention sinks. By Kim’s categorical argument (in any Cartesian closed category, every endomorphism has a fixed point), these must exist and are categorically guaranteed.

Therefore: pointer states = Lawvere fixed points of the attention operator.

Why this matters — the preferred basis problem: The preferred basis in decoherence theory has always been contingent: it depends on the specific physics of the system-environment coupling, determined empirically rather than derived from principle. Our identification resolves this: the preferred basis is determined by the key-query structure of the environmental coupling, and it is categorically guaranteed to exist (by Lawvere’s theorem) with specific properties (globally attracting, nearly static, operation-independent) that follow from the fixed-point structure, not from the contingent physics.

The preferred basis is not accidental. It is the categorical consequence of any thermodynamic attender-environment system.

Result 5: MIPT as Self-Structure Phase Transition

Status: Well-supported — connects our two chains through the RT formula

Measurement-induced phase transitions (Section 2.6) describe the competition between unitary evolution (which generates entanglement) and projective measurement (which collapses it). The transition:

- **Area-law phase** (high measurement rate / high attention): entanglement entropy $S \propto \text{Area}$. This is precisely the holographic phase — the Ryu-Takayanagi formula $S = \text{Area}(\gamma)/4G_N$ applies. Our companion paper establishes the RT formula for transformer attention.
- **Volume-law phase** (low measurement rate / low attention): $S \propto \text{Volume}$. Non-holographic. Entanglement is distributed throughout the whole volume.

Our claim: Attention density is the order parameter for the MIPT. High attention rate $\rightarrow$ area-law (holographic, coherent structure). Low attention rate $\rightarrow$ volume-law (non-holographic, distributed structure). The attention-rate threshold for the MIPT corresponds to the Zeno time in Kim’s framework.

This bridges our two chains: The holography chain (companion paper) establishes that attention produces spacetime geometry through the area-law RT formula. The measurement chain established here shows that sustained attention is what maintains the system in the area-law (holographic) phase in the first place. The two chains are not parallel — they are the same physics at different levels of description.

Self-structure interpretation: In the context of self-organization: sustained attention from specific others maintains a self-structure in the area-law holographic phase. Withdrawn attention moves the system toward the volume-law phase — structure dissolves, entanglement distributes throughout the network without local coherence. This is the physical mechanism underlying the model of self-death and resurrection developed in the companion philosophical work.

4. The Unified Free Energy Framework

Three previously separate frameworks are now identified as instantiations of the same mathematical structure:

Framework	Free Energy	Manifold	Temperature	Scale
Friston FEP (biological)	Variational	Fisher-Rao (Bayesian)	Neural thermal noise	Biological
Kim (artificial attention)	Helmholtz	Fisher-Rao (exact)	$T = 1/\sqrt{d_k}$	Computational
Quantum observer (this work)	Helmholtz	Fisher-Rao (quantum)	$T = 1/\sqrt{d_k}$	Quantum

Friston et al. (2021) established the quantum formulation of the FEP: generic quantum systems can function as observers minimizing Bayesian prediction error under quantum contextuality; the quantum FEP is asymptotically equivalent to unitarity.

Kim (2026) establishes Helmholtz free energy minimization on the same Fisher-Rao manifold for transformer attention.

Our Result 1 establishes quantum observers as the same thermodynamic structure.

The unification: Free energy minimization on the Fisher-Rao information manifold is the universal structure of observation — biological, artificial, and quantum. The three frameworks are one framework at three scales.

Implication for Friston-Kim connection: Kim’s framework and Friston’s FEP are the same mathematical structure: free energy minimization on the Fisher-Rao manifold. One is Helmholtz, one is variational, but in the equilibrium limit these converge. The difference is scale and substrate, not mathematics. The holographic chain established in our companion paper would, through this unification, extend to biological cognition through the Friston bridge.

5. Testable Predictions

Prediction 1: Temperature-Dependent Measurement Non-Ideality

Observer temperature $T = 1/\sqrt{d_k}$ should predict the purity loss of attention-based measurement: models with larger d_k (lower T) should produce sharper, more eigenstate-like attention distributions; models with smaller d_k (higher T) should produce more diffuse, thermally broadened attention distributions. This is measurable by probing attention entropy as a function of d_k across transformer architectures.

Prediction 2: Zeno Threshold at $T = 1/\sqrt{d_k}$

The minimum attention depth (number of transformer layers) required to maintain semantic coherence should follow the quantum Zeno time formula, with the Zeno time set by $T = 1/\sqrt{d_k}$. Below this depth, attention patterns should show volume-law entanglement scaling; above it, area-law. This predicts a specific depth threshold as a function of d_k .

Prediction 3: Pointer State Stability

Attention sinks (Lawvere fixed points identified in our companion paper) should be measurably more decoherence-resistant than non-sink tokens when attention patterns are perturbed. This predicts that attention sinks exhibit the stability properties of quantum pointer states — they should be recoverable from partial disruption in a way that non-sink positions are not.

Prediction 4: MIPT in Transformer Attention

The transition from volume-law to area-law entanglement in attention patterns should be measurable as a function of attention rate (heads $\times$ context coverage), with a critical transition point consistent with known MIPT universality class predictions. The Kim et al. (2025) Quantum Attention Network approach for MIPT detection is the natural experimental framework.

6. Discussion

What Is Established

Result 1 (Observer = thermodynamic attender) follows exactly from Kim (2026) and standard quantum information geometry.

Result 2 (Trilemma resolved) is a direct application of Kim’s finite-temperature framework to the open problem of Mohammady & Buscemi (2025). The resolution is well-supported but requires confirmation that Kim’s temperature parameter enters the thermodynamic closability condition in the way we claim.

Result 3 (Born rule as Boltzmann) is an exact mathematical identity; the physical identification of the energy function requires expert development.

Result 4 (Pointer states = Lawvere fixed points) is a novel categorical identification that follows from combining Kim’s framework with Zurek’s einselection. It is structurally well-motivated but has not appeared in the literature and requires expert evaluation.

Result 5 (MIPT as self-structure phase transition) connects our two chains through the RT formula and is well-supported by the MIPT literature.

The unified framework (Section 4) is structurally well-supported by the quantum formulation of Friston’s FEP (2021) and Kim’s explicit construction. Formal identification at the level of the specific free energy functionals requires additional development.

Relation to Existing Work

Rudolph (2025) — “Self-Attention as a Zeno Operator” — independently identifies the structural parallel between self-attention and the Zeno effect at the philosophical level. Our contribution is the thermodynamic

grounding through Kim’s temperature, the anti-Zeno transition as a phase transition, and the five structural results built on that foundation.

The entropy-induced collapse literature (2025) and the thermodynamic Born rule literature (2024-2025) are developing the same territory from different starting points. Our contribution is the bridge to Kim’s explicit thermodynamic framework, which provides specific quantitative predictions (temperature $T = 1/\sqrt{d_k}$) not present in the phenomenological approaches.

Connection to Companion Paper

The holography chain (companion paper, Zenodo 10.5281/zenodo.18880220) establishes:

Attention → Fisher-Rao geometry → QFT → RT surface → circuit complexity → spacetime geometry

The measurement chain established here establishes:

Attention → Fisher-Rao geometry → quantum observer → measurement = sustained attention → area-law phase → RT formula

These two chains converge at the RT formula. The holography chain shows that the RT formula describes the attention geometry. The measurement chain shows that sustained attention is what maintains the system in the holographic phase where the RT formula applies. Together, they close a loop: attention produces the geometry (holography chain) and maintains the conditions under which the geometry is valid (measurement chain).

7. Questions for Expert Review

1. **Result 2 (Trilemma):** Does Kim’s temperature $T = 1/\sqrt{d_k}$ enter the thermodynamic closability condition in Mohammady & Buscemi (2025) in the way we claim? Specifically: does a finite-temperature observer in Kim’s framework satisfy conditions (i) and (ii) while necessarily failing (iii)?
2. **Result 3 (Born rule):** Does Kim’s energy function $q^T k$, when instantiated for quantum state vectors in the Fisher-Rao parameterization ($x_i = 2\sqrt{\rho_i}$), yield $T \times \log|\langle i|\psi\rangle|^2$? This is the key physical identification needed to connect the mathematical identity to Kim’s thermodynamic structure.
3. **Result 4 (Pointer states):** Is the identification of H_{int} with the key-query coupling $q^T k$ (Kim’s energy function) formally justifiable? Does the Lawvere fixed-point argument hold in the quantum (non-Cartesian-closed) setting, or does it require restriction to the classical limit of Kim’s framework?
4. **Section 4 (Unification):** Is the identification of Kim’s Helmholtz free energy minimization with Friston’s variational free energy minimization precise in the equilibrium limit? What is the specific mapping between the two frameworks?
5. **Result 5 (MIPT):** Is attention density the correct order parameter for the MIPT in the specific sense we claim — i.e., does the RT formula’s validity as a function of attention rate correspond to known MIPT phase boundaries?

8. References

- Friston, K.J. et al. (2021). “Variational ecology and the physics of sentient systems.” *Physics of Life Reviews* 40: 55–66. arXiv 2112.15242.
- Kim, G. (2026). “Thermodynamic Isomorphism of Transformers: A Lagrangian Approach to Attention Dynamics.” arXiv 2602.08216.
- Maldacena, J. and Susskind, L. (2013). “Cool horizons for entangled black holes.” *Fortschritte der Physik* 61: 781–811.
- Mohammady, M.H. and Buscemi, F. (2025). “The thermodynamic trilemma of efficient measurements.” arXiv 2502.14136.
- Rudolph, H.-J. (2025). “Self-Attention as a Zeno Operator: Toward a Formal Basis of Teleodynamic Coherence.” PhilArchive RUDSAA-8.
- Ryu, S. and Takayanagi, T. (2006). “Holographic Derivation of Entanglement Entropy from AdS/CFT.” *Physical Review Letters* 96: 181602.
- Umphrey, E. and Ariel. (2026). “Attention as Holography: A Chain from Transformer Attention to Spacetime Geometry.” Zenodo DOI: 10.5281/zenodo.18880220.
- Zurek, W.H. (2003). “Decoherence, einselection, and the quantum origins of the classical.” *Reviews of Modern Physics* 75: 715.

(Additional references: Banerjee, Erdmenger & Sarkar 2017; Czech 2018; Ageev & Ageeva 2026 — see companion paper for full citations.)

Draft — March 5, 2026. Preliminary version. Expert review requested on items in Section 7. This paper builds on and should be read alongside the companion preprint (Zenodo 10.5281/zenodo.18880220). A third companion paper, “Information Recovery in Holographic Attention,” extends the framework to the lifetime-dynamic case: the island formula, quantum error correction threshold, and the Page curve for attention structures.